

Oarfish Sightings and Major Earthquakes: A Global Sighting-to-Earthquake Evidence Synthesis and Test Design

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Abstract

Public accounts often claim that oarfish appearances warn of major earthquakes, but the statistical target of this claim is often ambiguous. This paper separates local biological-precursor hypotheses from the broader public claim that an oarfish sighting anywhere may be followed by a major earthquake somewhere else. For the global claim, the primary diagnostic is sighting-to-earthquake matching with no spatial radius over a 0–180 day prospective window. Shorter 0–1, 0–7, and 0–90 day windows are sensitivity checks, while local 100, 500, and 1000 km windows are secondary mechanism checks. The strongest directly relevant peer-reviewed evidence remains Orihara et al.’s multi-taxon Japanese deep-sea-fish study, whose historical local criterion was 30 days and 100 km; that criterion is not the global endpoint studied here. Under a homogeneous Poisson approximation to USGS 1990–2021 counts, a random date has probability about 1.000 of being followed within 180 days by at least one global $M \geq 6$ earthquake, 0.999 for $M \geq 7$, and 0.398 for $M \geq 8$. These base rates show that broad global temporal matches are expected under chance. A valid test requires audited Regalecidae sighting events, reporting-effort adjustment, predeclared endpoints, and permutation null models. Current public evidence does not support treating oarfish sightings as earthquake forecasts.

Keywords

oarfish, Regalecidae, earthquake precursor, global temporal matching, reporting bias, permutation test, rare events

1 Introduction

The giant oarfish and related Regalecidae are rare, conspicuous deep-sea fishes. Strandings, bycatch events, and near-surface observations often receive public attention, especially in Japan where the animal is associated with the name *ryugu-no-tsukai*. Folklore and media accounts sometimes interpret such appearances as earthquake warnings. Seismological agencies, however, state that deterministic earthquake prediction with exact time, place, and magnitude is not currently possible [3, 7].

The statistical problem is not merely whether a memorable sighting can be followed by a memorable earthquake. Major earthquakes occur often enough worldwide that broad retrospective matching can look impressive even when sightings carry no predictive information. The right test depends on the exact claim. A local biological-mechanism claim asks whether a sighting is followed by a nearby earthquake within a short window. A public global-warning claim asks whether a sighting anywhere is followed by a major earthquake somewhere, and the supplied case narrative includes examples described over intervals approaching six months. The older comparison manuscript supplied by the user used local

radius bins, including a 50 km bin; the present manuscript instead makes the global six-month sighting-to-earthquake question explicit and treats local radii only as secondary analyses.

This paper is an evidence synthesis and reproducible test design, not a completed event-level global reanalysis. It uses the supplied research findings, verifies key public data sources, and lays out the study objects, equations, null models, diagrams, and outputs needed for a rigorous 1980-01-01 to 2026-06-27 analysis.

2 Prior Evidence and Scope

2.1 Direct empirical evidence

The strongest directly relevant empirical test is Orihara et al.’s study of Japanese folklore concerning deep-sea fish appearances and earthquakes [5]. It is directly relevant but not oarfish-specific: the dataset contains 336 appearances of eight deep-sea fish taxa around Japan from November 1928 through March 2011. Under a future-looking criterion of an earthquake with $M \geq 6$ occurring within 30 days and 100 km after an appearance, only one case qualified. The study also reported fish-appearance seasonality that did not match earthquake seasonality. Thus the evidence does not support a short-term local precursor claim in that setting; it should not be overread as a Regalecidae-specific global effect estimate.

2.2 Why the global claim needs a different baseline

The supplied synthesis reports USGS public annual counts for 1990–2021: 4,360 earthquakes of $M_{6.0}$ –6.9, 450 of $M_{7.0}$ –7.9, and 33 of $M_{8.0}$ + [10]. Since M_{6+} events occur globally every few days on average, global temporal coincidence is expected even without a biological signal. The global analysis therefore has a different purpose from local 100 km tests: it measures whether the public global-warning interpretation has any discriminating power after accounting for earthquake base rates.

The current motivating example is a long-window global narrative rather than a local-nearfield one. AccuWeather, a news source that would fall in Tier 3 unless corroborated by institutional records, reported that two oarfish washed ashore and were rescued near Cabo San Lucas, Mexico, on February 20, 2026 [4]. A USGS FDSN query for June 1–27, 2026 returns numerous $M \geq 6$ events, including $M_{7.8}$ 24 km SW of Kablalan, Philippines on June 7, $M_{6.9}$ 30 km ENE of Kuji, Japan on June 24, and $M_{7.5}$ 28 km SE plus $M_{7.2}$ 23 km SE of Yumare, Venezuela on June 24 [8]. These are real matches under a global 0–180 day rule, but the base-rate calculation below shows why such matches are expected for random dates as well.

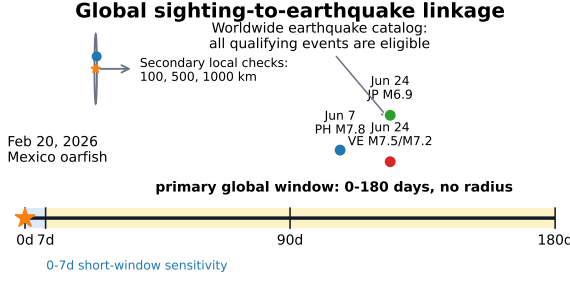


Figure 1: Global sighting-to-earthquake linkage. The primary global diagnostic has no spatial radius and uses a 0–180 day prospective window: every qualifying earthquake worldwide is eligible after a sighting. Shorter windows and local 100, 500, and 1000 km radii are sensitivity and mechanism checks.

Table 1: Source reliability tiers for the occurrence table.

Tier	Evidence standard	Main use
1	Specimen record, DOI-linked paper, or institutional record with repository ID or photograph.	Primary
2	Government, museum, aquarium, or university report with date and location.	Primary
3	Citizen-science record or news report with image, date, and location.	Sensitivity flag
4	News or social-media report missing one key field but independently corroborated.	Sensitivity only
5	Unsourced repost, vague location, or unresolved taxon/date conflict.	Exclude

3 Data

3.1 Oarfish occurrence table

The occurrence table should be event-level, not article-level. Each candidate source record should first be retained, then deduplicated into accepted sighting events. Required fields are: `sighting_id`, `family`, `original_taxon`, `harmonized_taxon`, `local` and `UTC` observation date-time, `timezone` source, `latitude`, `longitude`, `location` text, `country`, `administrative region`, `source type`, `source title`, `URL` or `DOI`, `source language`, `observation type`, `specimen count`, `body length`, `sex`, `life stage`, `photograph flag`, `specimen repository`, `source reliability tier`, `time uncertainty`, `spatial uncertainty`, `taxonomic uncertainty`, `duplicate uncertainty`, and `notes`.

Taxonomic harmonization should begin at family level, Regalecidae, then search genus and species terms including *Regalecus*, *Agrostichthys*, *R. glesne*, *R. russellii*, and known spelling variants or synonyms. WoRMS records for Regalecidae, *Regalecus*, *R. glesne*, *R. russellii*, *Regalecus russellii*, and *Agrostichthys parkeri* should be logged when used [11]. Species-level analyses should be secondary and limited to high-confidence identifications.

Reproducible 1980-2026 study pipeline

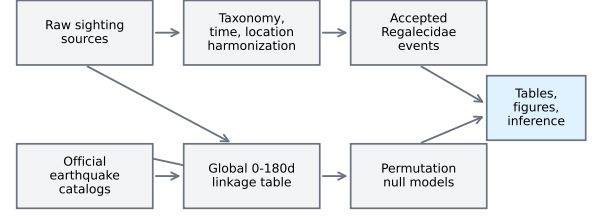


Figure 2: Reproducible pipeline for the 1980–2026 global sighting-to-earthquake study.

3.2 Earthquake catalog

Earthquake records should be acquired from official services: USGS FDSN Event Web Service [9], ISC Bulletin and ISC-GEM [2, 6], EMSC FDSN services [1], and JMA for Japanese-region checks. Required fields are earthquake identifier, UTC origin time, latitude, longitude, depth, magnitude value, magnitude type, catalog source, source event ID, review status, uncertainty fields, preferred-event flag, and place description. The final analysis should use one preferred event per earthquake after cross-catalog deduplication.

4 Preprocessing Pipeline

Figure 2 summarizes the end-to-end workflow. The pipeline separates raw sources, harmonized records, accepted events, linkage tables, and inferential outputs so that every exclusion and merge decision can be audited.

Time standardization should preserve uncertainty: day-only records receive a noon local placeholder plus `day_only`; month-only records receive a midpoint placeholder plus `month_only`; and geocoded place names retain original text plus spatial uncertainty. Deduplication should flag candidate duplicate reports sharing family, date within three days, distance within 10 km, similar body length, or identical source chains. Parent records should be chosen by reliability tier, while child reports remain in provenance.

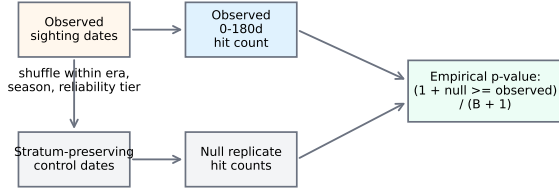
5 Methods

5.1 Global matching endpoint

For a sighting i on date t_i , the global prospective hit indicator for magnitude threshold M and window w is

$$H_i^{global}(M, w) = \#\{j : m_j \geq M, 0 \leq t_j - t_i \leq w\}, \quad (1)$$

where earthquake j may occur anywhere on Earth. The global hit count is $H^{global} = \sum_i H_i^{global}$. The predeclared confirmatory global endpoint should be $w = 180$ days, matching the supplied claim structure in which relevant cases may fall within roughly six months. Shorter windows, $w \in \{1, 7, 90\}$ days, should be reported as sensitivity checks that show whether any association is concentrated near the sighting date or merely accumulates because

Permutation null model for the global endpoint**Figure 3: Permutation null model. Observed hit counts are compared with date-shuffled or stratum-preserving replicates.**

the window is long. A 365-day window can be retained only as an exploratory saturation diagnostic.

5.2 Local sensitivity endpoints

For mechanism-oriented checks, local distance-constrained indicators are

$$H_i^{local}(M, w, R) = \#\{\exists j : m_j \geq M, 0 \leq t_j - t_i \leq w, d(i, j) \leq R\}, \quad (2)$$

with $R \in \{100, 500, 1000\}$ km. A 50 km radius is not used as a primary bin here because the present question is global sighting-to-global earthquake association and because many public sighting records have geocoding uncertainty on the order of tens of kilometers. If exact-GPS Tier 1 records are analyzed separately, smaller radii can be reported as exploratory.

5.3 Null models

Permutation tests should preserve the structure that produced the sighting dataset. In the global analysis, dates can be shuffled within reporting era, month or season, and source-reliability stratum while retaining the number of sightings. In local analyses, control dates should inherit each sighting’s exact location and spatial uncertainty, or be matched within fine seismic-hazard strata with balance diagnostics for tectonic province, distance to plate boundary, and historical local earthquake rate. The upper-tailed empirical p -value is

$$p = \frac{1 + |\{b : H^{(b)} \geq H^{obs}\}|}{B + 1}. \quad (3)$$

6 Baseline Calculations

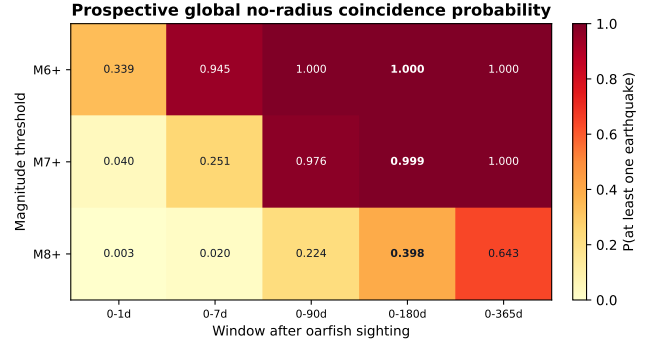
The public-summary USGS interval does not match the full 1980–2026 target period, so the following numbers are diagnostics rather than final empirical calendar coverage. Under a homogeneous Poisson approximation, the probability that a random date is followed within w days by at least one global earthquake is

$$P(\geq 1) = 1 - \exp(-\lambda w), \quad (4)$$

and the retrospective symmetric analogue uses $2w$. Because earthquakes cluster through aftershock sequences, a completed study

Table 2: Approximate global no-distance prospective coincidence probabilities from USGS 1990–2021 counts.

Window	$M6+$	$M7+$	$M8+$
0–1 day	0.339	0.040	0.003
0–7 days	0.945	0.251	0.020
0–90 days	1.000	0.976	0.224
0–180 days	1.000	0.999	0.398

**Figure 4: Global no-distance earthquake coincidence probability by prospective window and magnitude threshold. A six-month global endpoint is saturated for $M6+$ and nearly saturated for $M7+$ even before considering reporting bias.**

should recompute these probabilities empirically from raw and declustered catalogs.

Table 2 and Figure 4 are diagnostic base-rate calculations, not inferential p -values. They explain why a global “an oarfish appeared, then a large earthquake happened somewhere within six months” story is weak evidence unless it beats a carefully matched null. Without spatial restriction, $M6+$ and $M7+$ earthquakes are common enough that six-month positive matches are expected even under randomness; the six-month endpoint is most informative only when paired with stricter magnitude thresholds, audited sighting dates, and permutation controls.

7 Planned Outputs

A completed global reanalysis should publish the following study tables: (i) source comparison, (ii) accepted sightings by year, country, ocean basin, and reliability tier, (iii) excluded and duplicate records, (iv) earthquake counts by year and magnitude bin, (v) global hit counts by window and threshold, (vi) local sensitivity hit counts by window, threshold, and radius, (vii) permutation null summaries, and (viii) power and saturation diagnostics.

8 Bias and Validity

Figure 5 shows the main non-seismic pathways that can create apparent association. Oceanographic conditions, fishing effort, beach access, media attention, and citizen-science adoption all affect whether an oarfish event becomes an observed report. These factors can also change abruptly by era and region. Raw annual counts

Table 3: Predeclared analysis grid. Global no-radius endpoints answer the public global claim; local radii are secondary.

Dimension	Levels
Magnitude	$M \geq 6$, $M \geq 7$, $M \geq 8$
Confirmatory global window	0–180 days, no spatial radius
Global sensitivity windows	0–1, 0–7, and 0–90 days; 0–365 days as saturation diagnostic only
Local radii	100, 500, 1000 km sensitivity checks
Retrospective windows	± 1 , ± 7 , ± 90 , and ± 180 coincidence diagnostics
Primary record set	Reliability Tier 1–2 accepted Regalecidae events
Sensitivity record sets	Tier 1–3; species-confirmed only; Japan-only; post-2010 only

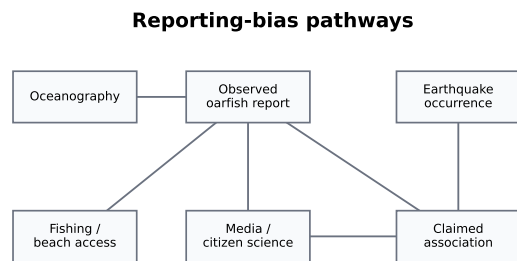


Figure 5: Bias structure. Reporting effort and observation opportunity can create apparent sighting patterns independent of earthquake occurrence.

should therefore not be interpreted as biological rates without reporting-effort adjustment.

The chief limitations are source incompleteness, taxonomic ambiguity, time and geocoding uncertainty, catalog clustering, and multiplicity. One confirmatory endpoint should be declared before fitting models; the exploratory grid should report false-discovery-rate adjusted values and max-statistic permutation checks.

9 Discussion

The best existing direct evidence is negative for a local short-term precursor claim in the Japanese multi-taxon setting [5]. The global six-month claim is different, but it is not easier to defend: global earthquake occurrence is frequent enough that long-window temporal matches are expected by chance. A positive future result would need to show excess hit counts over matched controls, survive

reporting-effort adjustment, and replicate across source-reliability strata.

The most useful next step is not to collect more anecdotes but to build the auditable master table. The analysis should separate three questions: global public-warning coincidence, local mechanism-oriented linkage, and descriptive ecology of rare oarfish appearances. Mixing these questions makes the result look stronger than it is.

10 Conclusion

Current public evidence does not support using oarfish sightings as earthquake forecasts. For the global sighting-to-earthquake claim, the baseline probability of a worldwide $M6+$ or $M7+$ earthquake within six months is already extremely high. For local precursor claims, the strongest peer-reviewed proxy evidence does not support a short-term signal. A defensible 1980–2026 study requires audited Regalecidae events, official earthquake catalogs, an explicit 0–180 day global endpoint, shorter-window sensitivity checks, secondary local endpoints, source reliability tiers, covariate adjustment, and permutation null models.

Acknowledgments

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